RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2018- 2019 (Odd Semester)

Innovative Practices Description

UNIT II: Embedded Computing Platform Design

Degree, Semester& Branch:VII Semester B.E. ECE B

Course Code & Title: EC6703 Embedded and Real Time Systems

Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic: CPU Bus - DMA, System Bus Configuration

Name of the Innovative Practice: Hardware Demonstration

Description:

Bus is a group of conducting wires which carries information, all the peripherals are connected to CPU through bus. A bus may be parallel or serial. Parallel buses transmit data across multiple wires. Serial buses transmit data in bit-serial format. A bus was originally an electrical parallel structure with conductors connected with identical or similar CPU pins, such as a 32-bit bus with 32 wires and 32 pins. The earliest buses, often termed electrical power buses or bus bars, were wire collections that connected peripheral devices and memory, with one bus designated for peripheral devices and another bus for memory. Each bus included separate instructions and distinct protocols and timing.

There are three types of buses:

1. Address bus - It is a group of conducting wires which carries address only. Address bus is unidirectional because data flow in one direction, from microprocessor to memory or from microprocessor to Input/output devices.

2. Data bus–It is a group of conducting wires which carries Data only. Data bus is bidirectional because data flow in both directions, from microprocessor to memory or Input/Output devices and from memory or Input/Output devices to microprocessor.

3. Control bus–It is a group of conducting wires, which is used to generate timing and control signals to control all the associated peripherals, microprocessor uses control bus to process data that is what to do with selected memory location.

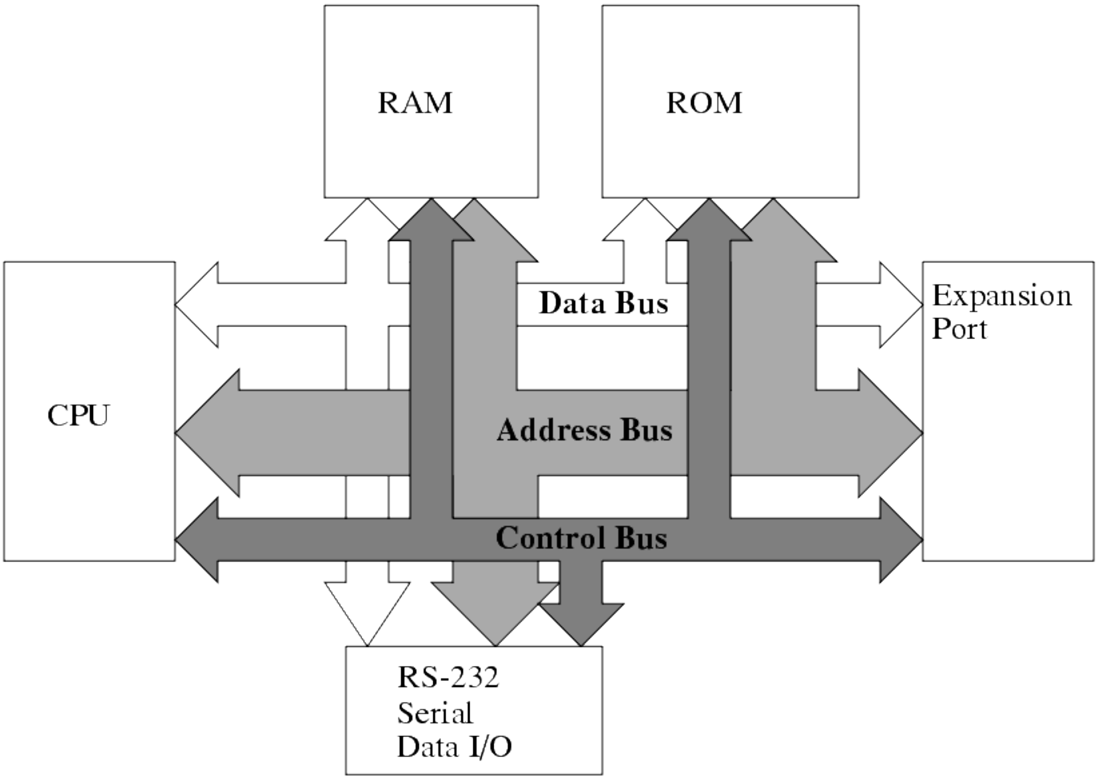

Figure: Peripherals interconnected with CPU

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	1	2	1	3	-	-	2	-	2	-	2

Reference :

- https://www.techopedia.com/definition/2162/bus
- http://engineeronadisk.com/notes_hardware/hardware5.html
- https://www.geeksforgeeks.org/bus-organization-of-8085-microprocessor
- Marilyn Wolf, “Computers as Components - Principles of Embedded Computing System Design”, Third Edition “Morgan aufmann Publisher,2012.